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Roll No

EX-8003(3)-CBGS)

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Special Machine

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) How dynamic torque is produced in switched Reluctance Motor.
b) Explain the advantages and disadvantages of Solid Rotors.
2. a) How to reduce the Torque Pulsations in Permanent magnet Synchronous Motor.
b) Explain any applications of Permanent Magnet Synchronous Motor.
3. a) Explain how load angle is controlled in Stepper Motor.
b) In detail explain sensor less control of Permanent Magnet Synchronous Motor.
4. Derive Torque Produced in Reluctance Motor.
5. Explain the PM synchronous motor construction and operation along with speed control and give its applications.

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6. Explain the construction and principle of operation of Brush Less DC Motor. What do you mean by winding pattern series and parallel.
7. Describe the VR and PM rotor structures and their torque production along with their torque angle characteristics.
8. Write the short note on any two of the following.
 - a) Design parameter of DC Brushed Motor
 - b) Speed control of DC Brushed motor
 - c) Permanent magnet material
